Aggregate functions and Grouping

Exercise 2 : ~~SQL fundamentals~~

Database : employees_db

id	first_name	last_name	department	Salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	chicago
10	Lauren	Hall	IT	50000	2020-08-10	San francisco

Questions

1) ~~Select statement~~ count() function

```
Select count(id) as number_of_employees
from employee_db;
```

number_of_employees
10

2) ~~AVG() function~~ Sum() function

```
Select sum(salary) as total_salary
from employee_db;
```

Total_salary
543 000

3) AVG() function

```
Select AVG(salary) as Average_salary, department
from employee_db
where department = "HR";
```

Averagee_salary	department
73 500	HR

4) Min() and Max function

```
Select min(salary) as lowest_salary,
max(salary) as highest_salary
from employee_db
```

lowest_salary	highest_salary
48 000	62 000

5) Group by statement

Select Distinct department, sum(salary) as total_salary

from employee_db

department	total_salary
IT	220 000
HR	99 000
finance	119 000
Marketing	105 000

6) ~~Grop~~ Group by and count()

select Distinct city, count(id) as total_employees

from employee_db

city	total_employees
New York	2
chicago	3
los ~~Angle~~ Angeles	2
San francisco	2
Houston	1

7) Group by and order by

Select Distinct department, Avg(salary) as average_salary

from employee_db

Group by department

Order by average_salary desc

~~department average salary~~
~~IT 5 500~~
~~HR 49 500~~
~~finance 54 600~~
~~Marketing 52 500~~

department	average_salary
finance	54 000
~~Marketing~~ IT	55 000
HR	49 500
~~IT~~ Marketing	53 000

8) Having clause

~~Select distinct department,~~

Select distinct department, sum(salary) as total_salary

from employee_db

~~where~~ Having total_salary < 100 00

Group by department

~~ID~~ department	total_salary
IT	220 000
Finance	119 000
Marketing	105 00

9) combining Group by, Having, Order by

Select ~~city~~ ~~distinct~~ city, count(ID) as number_of_employees.

from employee_db

Having number_of_employees < 1

Order by number_of_employees

Group by city

city	number_of_employees
chicago ~~New York~~	3
New York ~~chicago~~	2
los Angeles	2
San francisco	2

~~Houston~~

~~city number_of_employees~~

~~chicago~~

10) Combining Aggregate functions

```
select department , Avg(salary) as average_salary
from employee_db
  Group by department
  Order by average_salary desc
   limit 1
```

department	average_salary
finance	59 000